

Vitrine

An Agentic Pipeline for Volumetric
Cultural-Heritage Preservation

video ■ COLMAP ■ 3D Gaussian Splatting ■ SAM3 ■ depth-aware object
isolation
per-object mesh ■ native + composed USD scene graph

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Pairs with the upstream

LichtFeld Studio reconstruction engine

June 2026

Executive Summary

This is the **single current-state account** of **Vitrine**, the agentic pipeline that turns a hand-held walkthrough video of a museum or heritage exhibit into a structured, object-resolved 3D scene. It consolidates the three prior reports — the original bid (`main.tex`), the v2 architecture (`main_v2.tex`) and the v3 implementation plan (`main_v3.tex`), which remain as historical snapshots — and updates them with the **end-to-end validation performed in June 2026**.

The headline change since v3: the pipeline is no longer only *designed* — it has been **run end to end on a real reconstructed scene and now produces object-resolved, dual-USD output**. A single walkthrough now travels the full path: LichtFeld training of a Gaussian-splat field, SAM3 concept segmentation into named objects, a new depth-aware multi-view projection that isolates each object into its own gaussian subset and ranks them by curatorial significance, per-object meshing, and **two USD products** — a native LichtFeld splat USD and a textured composed multi-object scene graph with preview renders. Eight concrete defects that previously blocked this path were found and fixed, each surfaced by actually running the pipeline rather than by inspection.

Key Finding

Vitrine’s core promise — an *object-resolved* scene, not a monolithic splat — is now a working capability, validated on a real 80-frame indoor scene that resolved into five named objects (sculptures, furniture, walls, floor, ceiling), each isolated into its own gaussian subset and correctly ranked with the key items (sculptures, furniture) above the structural surfaces.

The honest boundary. This report uses an explicit **built & validated** / **in progress** / **designed** taxonomy and does not blur it. Validation was performed on a *reused* 80-frame reconstruction (`milo_run`), not on a fresh Google-Drive capture driven through ingest and COLMAP; the fresh-capture path needs Drive credentials and has not been exercised at scale. The per-object meshes are produced by the always-available `gsplat-to-TSDF` backend, *not* by the SOTA single-image hull generators (TRELLIS.2 / Hunyuan3D-2.1), whose ComfyUI custom-node dependencies are not yet built; and the generative *recovery* of unseen object faces (FLUX.2) and the local gemma-4 vision agent are staged but not wired into the loop. Those are reported as **in progress** or **designed** below, not as done. Quality, too, is first-pass: SAM3’s per-frame detection was sparse and the projection is multi-view-consistent but not yet depth-tested, so object masks are coherent but over-inclusive.

The report has four chapters: the project and its origin (§1); the architecture as it now stands (§2); the current implementation status, capability by capability, with the June 2026 validation results (§3); and what remains, with the decision-record status table and the path to a fresh-capture, SOTA-hull, fully-agentic run (§4).

Contents

Executive Summary	i
1 The Project and Its Origin	1
1.1 Why Vitrine exists	1
1.2 What the pipeline produces	1
2 Architecture as It Now Stands	2
2.1 One walkthrough, one structured scene	2
2.2 The two-agent separation	2
2.3 Operational scaffolding	2
3 Current Implementation Status	3
3.1 Capability status table	4
3.2 The validated end-to-end run	4
3.3 Defects found and fixed by running it	5
3.4 Known quality limitations	5
4 What Remains	6
4.1 Decision-record status	6
4.2 The path to a complete run	6

Chapter 1

The Project and Its Origin

1.1 Why Vitrine exists

Vitrine began as a University of Salford *AI Knowledge-Exchange Community of Practice* pilot, led from the MediaCity campus, partnered with DreamLab AI Consulting. The brief preserved in docs/brief/ named the problem plainly: the University has a growing catalogue of immersive, experiential, time-based arts (Nested Cinema, We Invented the Weekend, Lightwaves) but *no coherent or efficient pipeline for the preservation and emulation of site-based interactive and immersive experiences*. The pilot's aim was to test a novel, efficient pipeline that takes existing digital assets—video recordings, still imagery—and deploys *AI agentic workflows* to produce 3D objects suitable for emulation and preservation inside a games engine (the Salford MetaCampus, in Unreal).

1.2 What the pipeline produces

That framing maps directly onto what Vitrine builds. From one walkthrough video it produces:

- a photoreal **Gaussian-splat field** of the whole space (the high-fidelity record);
- a set of **named, individually-addressable objects**, each isolated from the field and placed at its measured pose (the curatorially-useful decomposition);
- a textured **polygonal environment** mesh and per-object meshes (the games-engine-ready geometry); and
- a single **USD scene graph** tying them together with source-video provenance—directly importable into Unreal or any USD-aware engine.

The work pairs with, and never forks, the upstream **LichtFeld Studio** reconstruction engine; Vitrine is the agentic orchestration, segmentation, decomposition and assembly layer on top of it.

Chapter 2

Architecture as It Now Stands

2.1 One walkthrough, one structured scene

The validated pipeline is a *reconstruct-then-decompose* flow, orchestrated by an in-container agent over the LichtFeld tool surface. The stages, in order:

```
video -> frames (+ per-image metadata sidecar)
      -> COLMAP SfM -> 3DGS training (LichtFeld: ImprovedGS+ / MRNF / MCMC)
      -> SAM3 concept segmentation -> key-item keyness ranking
      -> depth-aware multi-view mask -> 3D projection (per-object gaussian subsets)
      -> per-object meshing (gsplat-TSDF; CoMe / MILO backends interchangeable)
      -> native USD (LichtFeld convert) + composed textured USD (Blender)
```

2.2 The two-agent separation

A recurring design principle, unchanged from v3, is the clean separation between the *in-container orchestrator* (which decides what to run, calling LichtFeld and the pipeline stages as tools) and a *unified multimodal agent* (a vision-and-reasoning tool it calls for artefact judgement). There is no hidden state machine: the orchestration is explicit tool-calls over a manifest-driven configuration.

2.3 Operational scaffolding

Underneath sits the v3 operational layer, built and host-validated: a single human-authored `exhibit.toml` manifest with `env:` secret indirection and pre-snapshot stripping; a VRAM-bounded serial model lifecycle; a Docker service mesh (`v2g-net`) with service-DNS endpoints; an agent-controlled ComfyUI client; a state-of-the-art model registry with a host “idiot-check” wired into preflight; and a Rust/Axum web onboarding tool whose secret-containment is verified end-to-end.

Chapter 3

Current Implementation Status

This chapter is the heart of the report: each capability with its status and, where it was exercised, the **June 2026 validation result**. The validation scene throughout is `milo_run`: a real 80-frame indoor capture, reconstructed with COLMAP and trained to a 4M-gaussian splat (`splat_30000.ply`, LichtFeld `igs+` strategy, SH degree 3).

3.1 Capability status table

Capability	Status	Evidence / note (June 2026)
LichtFeld training in-container	built & validated	Runs after the CUDA-13 + vcpkg-OpenUSD + libz-ng runtime-lib fix; produced <code>splat_30000.ply</code> .
SAM3 concept segmentation	built & validated	5 objects (sculptures, furniture, walls, floor, ceiling) after patching the SAM 3.1 <code>addmm_act</code> bf16 dtype regression (sam3 #507).
Depth-aware object isolation (D10)	built & validated	New multi-view mask→3D projection: per-object gaussian subsets via COLMAP-camera voting. Isolated sculptures 1,080,171; furniture 452,401; floor 964,074; walls 357,276; ceiling 11,012 gaussians.
Key-item keyness ranking (FR-9)	built & validated	Curatorial priority × gaussian count ranks sculptures/furniture above structural surfaces despite floor’s larger count. <code>min_object_gaussians</code> threshold enforced.
Per-object meshing	built & validated	5 meshes via <code>gsplat-to-TSDF</code> after fixing the SH-degree PLY loader (the trained PLY is degree 1→3; was hardcoded to 45 coefficients).
Native USD export	built & validated	Via the LichtFeld CLI <code>convert</code> subcommand (headless; no GUI/MCP) → <code>scene_native.usd</code> (901 MB <code>splat</code> field).
Composed textured USD	built & validated	Blender assembler → <code>scene.usda</code> + 4 preview renders, after fixing the bake object-selection bug and the Blender 5.0 <code>usd_export</code> keyword change.
Manifest / lifecycle / service mesh	built & validated	<code>model_lifecycle</code> + <code>comfyui_control</code> suites 31/31; manifest loader + idiot-check host-validated (ADR-013).
SOTA model registry idiot-check	built & validated	Wired into preflight; full set (FLUX.2-dev, gemma-4-26B-A4B, TRELIS.2, Hunyuan3D-2.1, SAM3D) staged on the host.
Web onboarding tool	built & validated	Rust/Axum wizard; secret-containment verified; step-by-step credential instructions (ADR-015).
SOTA single-image hulls (TRELIS.2 / Hunyuan3D-2.1)	in progress	Weights staged; ComfyUI custom-node dependencies not built. Meshing falls back to <code>gsplat-TSDF</code> .
FLUX.2 generative recovery loop	in progress	Client built; not yet wired into the per-object hull path (unseen-face inpaint).
Local gemma-4 VLM serving	in progress	Q8_0 weights staged; <code>agent-vlm</code> serving not stood up.
v2g:* USD metadata + per-object composition	in progress	Supplied by the custom assembler; native-export <code>customData</code> parity probe pending.
Fresh-capture end-to-end (Drive→ingest→COLMAP)	designed	Validated on the reused <code>milo_run</code> scene only; Drive ingest needs credentials.
Automated acceptance gates + golden fixture	designed	27-gate harness and golden test scene specified, not yet built.

3.2 The validated end-to-end run

On `milo_run`, a single invocation now traverses segmentation → isolation → meshing → dual-USD assembly and produces: five isolated object PLYs; five per-object meshes; a 901 MB native

splat USD; and a composed, textured `scene.usda` with four preview renders. This is the first time the object-resolved promise has been demonstrated on real data rather than asserted.

3.3 Defects found and fixed by running it

The run was instructive precisely because each stage surfaced a real defect:

1. **SAM3 #507** — the SAM 3.1 `addmm_act` fused op casts to `bfloat16` and never restores, crashing segmentation; patched at the source binding.
2. **gsplat SH-degree** — the PLY loader hardcoded 45 spherical-harmonic coefficients but the trained field is SH degree 1 (9); now reads the actual degree and zero-pads.
3. **Hunyuan3D kwargs** — a `turbo` keyword crashed the client; kwargs now filtered to the constructor signature.
4. **LichtFeld runtime** — the binary needs the host CUDA-13 runtime, `vcpkg` OpenUSD libs and `libz-ng`; resolved via `LD_LIBRARY_PATH` (a Dockerfile fold-in remains).
5. **Depth-aware projection (D10)** — three sub-defects (empty-mask frames inflating the vote, absent SH vertex colours, and the world-space alignment) fixed to isolate objects correctly.
6. **Native USD path** — rewired from the never-running MCP server to the headless CLI `convert`.
7. **Blender bake** — the glTF-import `world` root was selected during bake; deselected.
8. **Blender USD export** — Blender 5.0 dropped a `usd_export` keyword; kwargs now filtered to the operator's RNA properties.

3.4 Known quality limitations

The isolation mechanism works; its quality is first-pass. SAM3 detected each object in only a minority of sampled frames, so the multi-view vote runs over few views; the projection is multi-view-consistent (and parallax filters gross background) but is *not* yet depth-tested per view, so object subsets are coherent but over-inclusive. Tightening this more sampled frames, per-view depth occlusion, and denser SAM3 prompts is the first quality task, distinct from the functional milestone reported above.

Chapter 4

What Remains

4.1 Decision-record status

ADR	Subject	Status
009	Per-video ingest; per-image metadata sidecar; delete-on-verified-extraction	in progress
010	Key-item ranking; depth-aware isolation; hull recovery; pose persistence	built & validated / in progress
011	USD v2g:* metadata; native + composed USD; lineage	built & validated / in progress
012	SOTA tooling: CoMe mesh, ALIKED+LightGlue, TRELIS.2/Hunyuan, FLUX.2, pins	in progress
013	exhibit.toml manifest; serial lifecycle; v2g-net; gemma-4 agent	built & validated
014	Agent-controlled ComfyUI; native USD via CLI convert; recovery loop	built & validated / in progress
015	Schema-driven web onboarding; hardware-aware selection; Drive write-back	built & validated

built & validated = built and exercised on real data **in progress** = core built, integration pending **designed** = specified, build pending

4.2 The path to a complete run

In rough dependency order, the work that turns the validated milestone into a production capability:

1. **Quality:** densify SAM3 sampling and add per-view depth occlusion to the D10 projection.
2. **SOTA hulls:** build the TRELIS.2 / Hunyuan3D-2.1 ComfyUI custom-node dependencies, replacing the gsplat-TSDF fallback for key objects.
3. **Recovery:** wire the FLUX.2 unseen-face inpaint loop and stand up the local gemma-4 VLM as the verify-or-retry judge.
4. **Metadata:** populate the v2g:* per-object USD customData and run the native-export parity probe; persist per-object pose.
5. **Fresh capture:** exercise the Drive→ingest→COLMAP→... path end to end on a new capture with credentials.
6. **Assurance:** author the golden test fixture and automate the acceptance gates; fold the libz-ng runtime lib and host-staged weights into the container image.

Recommendation

Vitrine has crossed the line from *designed* to *demonstrated*: a real capture now becomes an object-resolved, dual-USD scene. The next increment of value is *quality and SOTA hulls* on the same validated path, followed by a fresh-capture run — not new architecture. The honest gaps above are the backlog, in priority order.